

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Computer Applications in Geology L-T-P:3 – 0 – 0
Credits:3

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

Nature of Geological data – field, experimental, analytical and digital images – standard techniques of analyses and presentation of these data using a Computer – overview of software platforms implementing them; Multivariate data in Geology and analysis using standard linear algebraic techniques – Principal Components, Factor, Cluster and Discriminant Analyses; Solution of systems of nonlinear equations using Newton-Raphson technique and application to understanding of Chemical Equilibria in general and fluid-rock interaction in particular, acquaintance with software implementations (such as PHREEQC and Geochemists Work Bench) ; Concepts of analysis of spatial and temporal data in Geology; Introduction to Python programming for solving Geological problems.

Reference Books:

- 1) Numerical Analysis for Engineers by Steven Chapra and Ramond Canale (Mc Graw Hill)
- 2) MATLAB Recipes for Earth Sciences by Martin Trauth (Elsevier)
- 3) Introduction to Python in Earth Science Data Analysis – from descriptive Statistics to Machine Learning by M. Petrelli (Elsevier)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction to types of Geological Data and overview of methods of Analysis and presentation	6
2.	Multivariate Statistics and Applications in Geology	9
3.	Nonlinear systems and applications to Fluid-rock interaction problems	6
4.	Analysis of Spatial and Temporal Data in Geology	9
5	Python programming in solving Geological problems	6
Total number of hours		36

